

Obliquity Change from a Planetary Encounter

Eric Pfahl — KITP — 04/28/06

Consider two planets whose orbital radii from the Sun differ by a small amount s from the mean radius r . We suppose that s is larger than the respective Hill spheres of the planets. The gravitational interaction between the planets is largest near inferior conjunction, when the mutual separation is $\sim s$. Since the planets have a relative orbital speed of $\delta v \simeq \Omega_r s$, the timescale over which the planets interact is $s/\delta v \sim \Omega_r^{-1}$, where $\Omega_r \simeq (5 \text{ yr})^{-1} (r/10 \text{ AU})^{-3/2}$ is the heliocentric Keplerian frequency at radius r .

Each planet has a finite spin-induced oblateness, for which the lowest order contribution to the noncentral exterior potential at distance s is

$$\delta V = J_2 \frac{GM}{R} \left(\frac{R}{s} \right)^3 P_2(\theta) , \quad (1)$$

where M and R are the mass and equatorial radius of the planet, $P_2(\theta)$ is the second Legendre polynomial, and J_2 is a dimensionless weight. When a second body approaches the oblate planet, it exerts a specific torque of $\partial \delta V / \partial \theta$. Letting the planets have labels A and B , we see that the torque on planet A due to B at inferior conjunction is, up to a trigonometric factor,

$$N_{AB} \sim M_B J_{2A} \frac{GM_A}{R_A} \left(\frac{R_A}{s} \right)^3 . \quad (2)$$

The effect of the torque is to cause the spin vector of A to precess at a rate of order $N_{AB}/(I_A \omega_A)$, where $I_A = k_A M_A^2 R_A^2$ and ω_A are the planet's moment of inertia and spin frequency. Over the characteristic interaction time, the angle $\delta \epsilon$ through which the spin axis migrates is

$$\delta \epsilon \sim \frac{J_{2A}}{k_A} \frac{GM_B}{s^3 \omega_A \Omega_r} \simeq 5 \times 10^{-4} \left(\frac{J_{2A}}{0.01} \frac{0.1}{k_A} \frac{M_B}{M_J} \frac{T_A}{1 \text{ day}} \right) \left(\frac{1 \text{ AU}}{s} \right)^3 \left(\frac{r}{10 \text{ AU}} \right)^{3/2} \text{ deg} , \quad (3)$$

where M_J is the mass of Jupiter, and T_A is the planet's rotation period. As a particular example, we estimate the obliquity change of Uranus due to an encounter with Neptune. The relevant physical data for Uranus are $J_2 \simeq 3 \times 10^{-3}$, $k \simeq 0.2$, and $T \simeq 0.7 \text{ day}$, while the mass of Neptune is $\simeq 0.05 M_J$. For a single close encounter with $s = 0.2 \text{ AU}$ at $r = 10 \text{ AU}$, the obliquity change is $\delta \epsilon \sim \text{few} \times 10^{-4} \text{ deg}$.